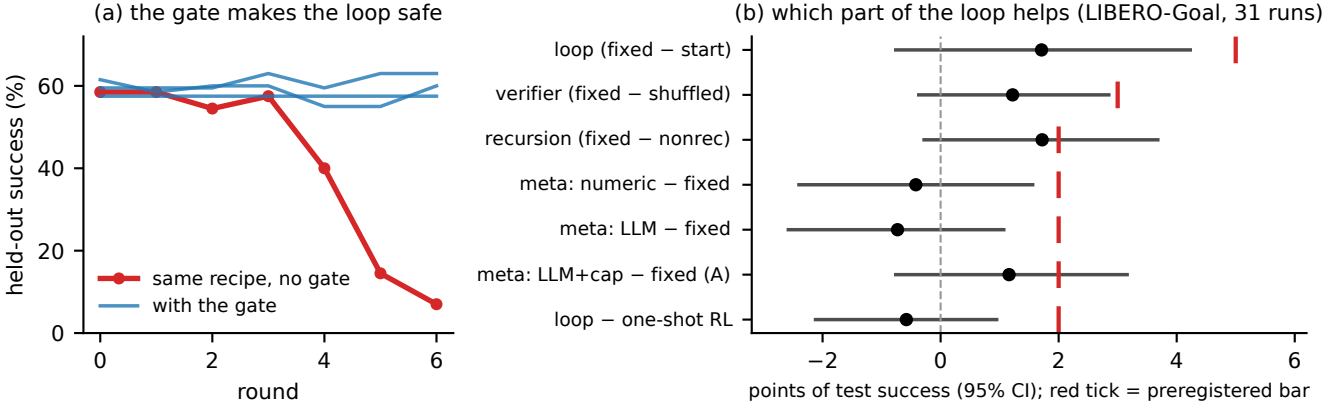
(a) the gate makes the loop safe
held-out success (%)
0
20
40
60
0
2
4
6
round
same recipe, no gate
with the gate
(b) which part of the loop helps (LIBERO-Goal, 31 runs)
loop (fixed − start)
verifier (fixed − shuffled)
recursion (fixed − nonrec)
meta: numeric − fixed
meta: LLM − fixed
meta: LLM+cap − fixed (A)
loop − one-shot RL
−2
0
2
4
6
points of test success (95% CI); red tick = preregistered bar